Infosys Springboard

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Basic Git Commands

1. git add
Adds all the changed files in the GitHub

2. git clone https
Downloads a project from GitHub to your computer.

3. git pull
Gets the latest changes from GitHub AND adds them to your project.

4. git fetch --all
Gets the latest changes from GitHub but does not touch your files. Just updates Git's info.

5. git branch --all
Shows all branches like your branches and GitHub branches combined.

6. git branch
Shows only your local branches.

7. git checkout branchname
Moves you to that branch.

8. git checkout -b branchname
Makes a new branch AND moves you to it.

9. git add .
Adds all changed files at once.

10. git commit -m "message"
Saves your changes with a small message explaining what you did.

11. git push
Sends your changes to GitHub.

1. Git
Git is a tool installed on your computer that tracks changes in code.
It lets you save versions, undo mistakes, and manage branches.

2. GitHub
GitHub is an online website where you store Git projects and collaborate with others.
It hosts your Git repositories on the cloud.

3. GitLab

GitLab is also an online Git repository platform, but it includes extra DevOps features like CI/CD pipelines.
It is similar to GitHub but more automation-focused.

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SDLC (Software Development Life Cycle) – Short Notes

SDLC is the step-by-step process of designing, developing, testing, deploying, and maintaining software.

SDLC Roles

1. Business Analyst / Functional Consultant

- Collects requirements
- Prepares Functional Specification Document (FSD)

2. Developer / Programmer

- Writes code
- Prepares Technical Specification Document (TSD)
- Creates POC and does Unit Testing

3. Tester / QA

- Tests the software
- Prepares Test Scripts & Test Results

Main Servers: Development, QA/Quality, Production

4. Admin Team – Handles servers & deployments
5. Project/Product Management – Plans & monitors project
6. Sales & Marketing – Promotes product
7. HR – Hiring & onboarding
8. Documentation Team – User manuals & guides
9. Support Team – Post-deployment issue handling

Waterfall Model

A linear model:
Requirements → Design → Development → Testing → Deployment → Maintenance

Agile Methodology

An iterative model with small cycles called Sprints.

Agile Roles:

- Product Owner
- Scrum Master
- Development Team

Common SDLC Steps

Design → Develop → Test → Deploy

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API

API (Application Programming Interface)
A way for the frontend and backend to communicate.

- Frontend sends payload (data)
- Backend receives payload, processes it
- Backend sends response back to frontend

HTTP Methods

- GET – fetch data
- POST – send data (payload/body)
- DELETE – remove data

Flask API – Basics

```
from flask import Flask, request, jsonify

app = Flask(__name__)
```

- Flask → creates the API application
- request → receives payload
- jsonify → sends response

FastAPI – Basics

from fastapi import FastAPI

app = FastAPI()
FastAPI Endpoints

@app.get("/")
@app.post("/send")

- FastAPI supports async functions (faster & modern)

Postman

A tool used to test API endpoints (GET, POST, DELETE) by sending HTTPS requests.

WebSocket

A connection that stays open for real-time communication (chat apps, live updates).

Key Idea

API = Frontend → (payload) → Backend → (response) → Frontend

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SYSTEM OVERVIEW – AUTHENTICATION AND DASHBOARD ARCHITECTURE

1. LOGIN / REGISTRATION PAGE

Registration Form Fields:

- Name
- Email
- Password
- Preferred Language

Login Form Fields:

- Email / Username
- Password
- Login with Google

The Login and Registration page allows users to securely create an account and access the system using either traditional login credentials or Google authentication.

2. BACKEND (API LAYER)

The backend is developed using FastAPI.

Responsibilities of the Backend:

- User registration
- User login
- Google login integration
- File upload handling
- Dashboard API management

Security Measures:

- JWT (JSON Web Token) is used for authentication.
- Password hashing is implemented to ensure passwords are never stored in plain text.

3. PASSWORD HANDLING PROCESS

The password handling process ensures secure authentication:

1. The user enters their password.
2. The password is hashed using a secure hashing algorithm.
3. The hashed password is stored in the database.
4. During login, the entered password is hashed again.
5. The system compares the hashed values for verification.

This ensures strong protection of user credentials.

4. DATABASE OPTIONS

The system can use any of the following databases:

- SQLite (suitable for small projects)
- MySQL
- PostgreSQL
- MongoDB

The choice of database depends on project size and scalability requirements.

5. USER TABLE STRUCTURE (Example)

Column Name: user_id
Description: Unique user identification number

Column Name: username
Description: User’s display name

Column Name: email
Description: User’s registered email address

Column Name: password
Description: Hashed password stored securely

6. GOOGLE LOGIN

Google login is implemented using the Google OAuth API.

Features:

- Users can log in using their Google account.
- No password is required within the system.
- Google provides verified user details for authentication.

This improves convenience and security.

7. DASHBOARD PAGE

After successful authentication:

- The user is redirected to the Dashboard page.

Dashboard Features:

- File upload functionality
- View uploaded files

8. FILE UPLOAD

Supported File Types:

- PDF
- Images
- Scanned documents

Files are securely stored on the server or cloud storage for further processing.

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Database & Authentication (Condensed Notes)

1. CRUD Operations

- Create → Add a new table or record
- Read → Display data
- Update → Modify existing data
- Delete → Remove data

2. Basic SQL Commands

CREATE

Creates a table.

```
CREATE TABLE User (id INT, username VARCHAR(60), password VARCHAR(60));
```

INSERT

Adds new data.

```
INSERT INTO User VALUES (1, 'sriya', 'password123');
```

SELECT

Reads data.

```
SELECT * FROM User;
```

UPDATE

Modifies data.

UPDATE User SET username='admin' WHERE id=1;
DELETE

Removes data.

DELETE FROM User WHERE id=1;

3. DROP vs TRUNCATE vs RENAME

DROP

- Deletes entire table/database
- Removes structure + data

TRUNCATE

- Deletes only rows
- Keeps structure

RENAME

Rename a table.

4. SQL Operators

Comparison Operators

- =
- !=
-
- <

Logical Operators

- AND
- OR
- NOT

Example:

SELECT * FROM User WHERE id=1 AND username='sriya';

5. ORDER BY & GROUP BY

ORDER BY

Sorts results.

SELECT * FROM User ORDER BY username ASC;
GROUP BY

Groups rows.

SELECT department, COUNT(*) FROM Employee GROUP BY department;

6. Aggregate Functions

- MIN()
- MAX()
- AVG()
- COUNT()

Example:

SELECT COUNT(*) FROM User;

7. SQL Pattern Matching (LIKE)

- A% → starts with A
- %a → ends with a
- %a% → contains a

8. SQL Joins

- INNER JOIN → Only matching rows
- LEFT JOIN → All left table rows + matches
- RIGHT JOIN → All right table rows + matches
- CROSS JOIN → All possible combinations

9. Special Keywords

- IN → multiple values
- BETWEEN → range

- LIKE → pattern search
- IS NULL → null check

10. UNION Commands

- UNION → removes duplicates
- UNION ALL → keeps duplicates
- INTERSECT → common rows
- EXCEPT → rows in first not in second

11. HAVING Clause

Used after GROUP BY to filter groups.

12. JWT (JSON Web Token)

What is JWT?

A secure token format used for:

- Authentication
- Authorization
- Stateless APIs

JWT Flow

1. User logs in
2. Server validates credentials
3. Server generates JWT
4. Client stores token
5. Client sends token in header
6. Server verifies token and grants access

Example in Python

```
import jwt, datetime
SECRET_KEY = "mysecretkey"
payload = {
```

```
  "user_id": 101,
  "username": "sriya",
  "exp": datetime.datetime.utcnow() + datetime.timedelta(minutes=30)
}
token = jwt.encode(payload, SECRET_KEY, algorithm="HS256")
```

13. OAuth (Overview)

OAuth allows secure login without sharing passwords.

Examples:

- Login with Google
- Login with GitHub
- Login with Facebook

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HASHING AND PYTHON CONCEPTS

1. HASHING

Hashing is a technique used to protect sensitive data such as passwords.

What is Hashing?

- Hashing converts a password into a fixed-length, unreadable value.
- It uses a mathematical algorithm.
- Salting adds extra security to the hashing process.
- The hash value cannot be converted back to the original password.

Hashing ensures that even if the database is compromised, actual passwords are not exposed.

2. BCRYPT HASHING (PYTHON)

bcrypt is a Python library used for secure password hashing.

Installation:

```
pip install bcrypt
```

Import:

```
import bcrypt
```

Hashing Example:

```
import bcrypt
password = "rithwik".encode()
salt = bcrypt.gensalt()
hashed = bcrypt.hashpw(password, salt)
print(hashed)
```

Password Verification Example:

```
entered_password = "rithwik".encode()
if bcrypt.checkpw(entered_password, hashed):
print("Password matched")
else:
print("Invalid password")
```

In this process:

- The password is encoded into bytes.
- A salt is generated automatically.
- The password is hashed securely.
- During login, the entered password is verified against the stored hash.

3. PYTHON STRING BUILT-IN METHODS

strip()
Removes spaces from both sides of a string.

Example:
```
text = " pardhu "
print(text.strip())
```

join()
Joins elements using a separator.

Example:
```
letters = ["g", "f", "k"]
print("".join(letters))
 : gfk
```

replace()
Replaces part of a string with another value.

Example:
```
text = "I like python"
print(text.replace("python", "java"))
```

startswith()
Checks if a string starts with specific characters. Returns True or False.

endswith()
Checks if a string ends with specific characters. Returns True or False.

find()
Returns the index position of a substring.

isdigit()
Checks if the string contains only numbers.

Example:
"1235".isdigit()

isalpha()
Checks if the string contains only letters.

4. DICTIONARY METHODS

get()
Returns the value for a given key.

keys()
Returns all keys in the dictionary.

values()
Returns all values in the dictionary.

items()
Returns key-value pairs.

pop("Age")
Removes a specific key.

popitem()
Removes the last inserted key-value pair.

clear()
Removes all items from the dictionary.

5. SET METHODS

add()
Adds an element to the set.

remove()
Removes an element. Gives an error if the element does not exist.

discard()
Removes an element safely (no error if missing).

pop()
Removes a random element.

clear()
Removes all elements from the set.

union()
Combines two sets.

intersection()
Returns common elements between sets.

difference()
Returns elements in one set but not in another.

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Milestone 1

Date:13/2/2026

Class methods

getattr(obj, name[, default]): Returns the value of a named attribute from an object, optionally returning a default value if not found.

setattr(obj, name, value): Assigns a new value to a named attribute of an object, creating the attribute if it does not exist.

hasattr(obj, name): Returns True if the object has the specified attribute, otherwise returns False.

delattr(obj, name): Deletes a named attribute from an object, equivalent to the del statement.

isinstance(obj, classinfo): Checks if an object is an instance of a specified class or a tuple of classes, returning True or False.

Issubclass: checks if a class is a subclass of another class (or a tuple of classes).

Miscellaneous functions

globals() → Returns a dictionary of the current global symbol table.

locals() → Returns a dictionary of the current local symbol table.

callable(obj) → Returns True if the object can be called like a function, else False.

eval(expr) → Evaluates a Python expression (string) and returns its result.

exec(code) → Executes Python statements dynamically from a string or code object.

Exception handling

try → Write the code that might cause an error

except → What to do if an error happens (like catch)

finally → Code that always runs, error or no error

ex:

```
try:
    x = int(input("Enter a number: "))
    print(10 / x)
except:
    print("Error occurred (maybe division by zero or wrong input)"
finally:
    print("Program ended")
```

o/p:

if x==0

Error occurred (maybe division by zero or wrong input)

Programm ended

Assignment – 3

Enumerate:

enumerate() is used when you loop through something and want both

the index (position)

and the value

at the same time.

Use it when we are looping through lists, strings, tuples

Without enumerate	With enumerate
1.range(len(x))	1.enumerate(x)
2.More code	2.Cleaner
3.Error-prone	3.Safe

:

isinstance ():

isinstance() checks whether an object belongs to a specified class or data type and returns True or False.

Syntax: isinstance(object, classinfo)

object → variable / value

classinfo → data type or tuple of types

Returns True or False

Ex:

1)x = 10

print(isinstance(x, int))

o/p: True

2) class Student:

```
    pass
```

```
s = Student()
```

```
print(isinstance(s, Student))
```
o/p=> True

GClibrary:

1) The gc library means Garbage Collection.

2) gc automatically cleans memory by removing objects that are no longer needed.

3) gc frees that unused memory

:

Date 14/2/2026

AI models:

They are programmed and pretrained for specific purposes ,it can be chat ,image generation,prediction etc..

Prediction -> weather prediction

Image generation -> open cv(computer vision)

Platforms:

Company	Main LLM
Google	Gemini
OpenAI	GPT
Microsoft	Copilot (GPT-based)
Anthropic	Claude
Meta	LLaMA
Amazon	Bedrock (multi-model)

LLM(large language models):

Large Language Models are AI systems trained on massive text datasets to understand, generate, summarize, translate, and reason with language. They learn patterns in text and predict the next token (piece of text).

Token limit :maximum number of tokens (input +) a model can handle in one request

Rate limit : how often and how much you can call an LLM within a given time window.

Chat completion component for open ai:

response=client.chat.completions.create(model: “gpt-4o-mini”,message=messages ,temperature=0.7,max-tokens=500)

Temp-> 0-1.1

creativity of model will be used with high temp;

creating api :

1)Open ai api

2)Gemini api

3)Lama api

Day15:

NLP:

it’s a subset of ai teaching computer to read and understand languages

two types of sentences : 1)positive sentence

2)negative sentence

NLTK:(Natural language tool kit)

It’s a python library used to interpret techniques

=>pip install nltk

=>import nltk

=>nltk.download(‘all’)

7main key features:

1) Tokenization:

Tokenisation is the process of breaking text into smaller pieces called tokens so that a language model can understand and process it.

=>from nltk.tokenize import word_tokenize

2) stop word removal :

Removing stop words like is ,are,in,and

These are words that doesn’t have any meaning

This->not a stop word

=> from nltk.corpus import stopwords

3) Stemming:

It is the process of reducing a word to its root (base) form by removing prefixes or suffixes.

The root word may not be a real dictionary word.

=>from nltk.stem import poster Stemmer

4) Lemmatization:

Lemmatization is the process of converting a word into its base or dictionary form, called a lemma.

=>from nltk.stem import Word Net Lemmatizer

5) parts of speech tagging ->pos tagging

POS Tagging means labelling each word in a sentence with its grammatical role

(like noun, verb, adjective, etc.).

=>from nltk.tokenize import word_tokenize

6)NER (Name Entity Recognition)

Recognise names , place, data,organisations

=>nltk.download('maxent_ne_chunker')

7)Text preprocessing

Text preprocessing means cleaning and preparing raw text so that machines can understand it better.

ASSIGNMENT 4

:

Date:15/2/2026

Machine learning:

Machine Learning (ML) is a field of AI where computers learn patterns from data and make predictions without being explicitly programmed.

Types of Machine Learning

Supervised Learning

Uses labeled data (input +).

Examples: spam detection, price prediction.

Unsupervised Learning

Uses unlabeled data to find patterns.

Examples: clustering customers, grouping data.

Reinforcement Learning

Learns using rewards and penalties.

Examples: games, robotics, self-driving cars.

Embedding

Embedding is a numerical vector representation of data (text, image, etc.) that captures semantic meaning.

Underfitting

Model is too simple and fails to learn patterns.

Low training and testing accuracy.

Overfitting

Model learns noise and memorizes data.

High training accuracy, low testing accuracy.

Logistic regression:

used to make YES/NO decisions.

1.Looks at input data (marks, age, hours studied, etc.)

2.Calculates a score

3.Converts that score into a probability (0 to 1)

4.Decides:

oYES (1) if probability ≥ 0.5

oNO (0) if probability < 0.5

Why is it called logistic?

Because it uses a special S-shaped curve called the sigmoid function.

This curve:

Never goes below 0

Never goes above 1

Perfect for probability

Linear regression:

method used to predict a NUMBER.

1.Looks at input (hours studied, experience, area, etc.)

2.Draws the best straight line through the data

3.Uses that line to predict a number

Linera regression because the relationship looks like a straight line:

y = mx + c

sklearn and sckit-learn:

scikit-learn → full name of the machine learning library

sklearn → short name (nickname) used in code

import scikit-learn ❌ (not allowed)

import sklearn ✅

pip install scikit-learn ✅

pip install sklearn ❌

Python module names cannot contain hyphens (-).so sklearn instead of scikit-learn

scikit-learn (sklearn) does not provide pretrained models; it provides algorithms that must be trained on your own data.

Decision trees:

makes decisions by asking simple YES/NO questions, like a flowchart.

It works like this:

1.Ask a question

2.Go left or right based on the answer

3.Ask another question

4.Reach a final decision

OCR models:

1.PaddleOCR is a deep-learning-based OCR system developed by Baidu. It is known for high accuracy and strong performance on real-world documents like bills, ID cards, signboards, and forms. It supports many languages, including English and Indian languages, and can handle printed text, some handwriting, and complex layouts.

2.PyTesseract OCR is a Python wrapper for the Tesseract OCR engine. It works best for clean, simple, printed text such as scanned books or documents with good lighting and contrast. It is easy to use but not very accurate for handwriting or noisy images.

3.DocTR (Document Text Recognition) is a modern OCR system based on deep learning. It not only extracts text but also understands document structure, such as lines, blocks, and paragraphs. This makes it useful for advanced document analysis, research, and layout-aware OCR tasks.

4.Surya OCR is mainly focused on Indian languages and regional scripts. It is useful when working with documents in languages like Telugu, Hindi, Tamil, etc. It is often chosen for local-language digitization projects in India.

5.EasyOCR is beginner-friendly and very easy to set up. It supports many languages and works well for quick OCR tasks like reading text from images or screenshots. While it is simpler than PaddleOCR, it still gives good accuracy for general use cases.

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Steps:

1) Pip install pytesseract pillow

2)then download tesseract and save it in c drive in programming files and then copy the path

3)downalod an img of invoice save it in same folder in vscode

4) now in vs code use that path and code then copy path of img in vs code and paste it in py code

5)run python app.py

Code:

Invoice:

:

Date:17/2/2026

K-Means:

K-Means is a clustering algorithm.

It groups similar data points into K groups (clusters).

Here K means how many groups you want.

🔷 Simple idea

You choose a number K (for example, 3).

The algorithm puts data into 3 groups based on similarity.

Each group has a center called a centroid.

🔷 How K-Means works (step by step)

1.Choose K (number of clusters).

2.Pick K random points as starting centers (centroids).

3.Assign each data point to the nearest centroid.

4.Move the centroid to the average (mean) of its group.

5.Repeat steps 3–4 until clusters stop changing.

Euclidean Distance

Measures how far a data point is from a centroid

K-Means assigns a point to the nearest centroid

Most commonly used distance metric

📐 Formula:

x2−x1)2+(y2−y1)2

👉 Smaller distance → point is more similar

Within-Cluster Sum of Squares (WCSS)

Measures how compact a cluster is

It is the sum of squared distances between each point and its centroid

Interpretation of WCSS

Large WCSS → clusters are more spread out

Small WCSS → clusters are tight and compact

Goal of K-Means: minimize WCSS

👉 Used in the Elbow Method to choose the best value of K

Elbow Method

The Elbow Method is used to find the best value of K (number of clusters) in K-Means.

Basic idea

Run K-Means for different values of K (1, 2, 3, …)

For each K, calculate WCSS (Within-Cluster Sum of Squares)

Plot K vs WCSS

Look for a point where the graph bends like an elbow

👉 That point is the optimal K

KNN

K-Nearest Neighbours (KNN) is a supervised learning algorithm.

It predicts the based on the K closest data points (neighbors).

K = number of nearest neighbors

Uses distance (usually Euclidean distance)

KNN for Classification

Used when the is a category (label).

How it works

1.Choose value of K

2.Find K nearest data points

3.Take a majority vote

4.Assign the class with maximum votes

KNN for Regression

Used when the is a number (continuous value).

How it works

1.Choose value of K

2.Find K nearest data points

3.Take the average of their values

4.Assign that average as prediction

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Back propagation

Backpropagation is an algorithm used to train neural networks.

It helps the network learn from mistakes by updating weights to reduce error.

Why is it needed?

A neural network makes a prediction

Prediction has an error

Backpropagation adjusts weights to minimize this error

How Backpropagation works

1️⃣ Forward Propagation

Input is passed through the network

is predicted

2️⃣ Loss Calculation

Compare predicted with actual

Error is calculated using a loss function

3️⃣ Backward Propagation

Error is sent from layer to input layer

Gradients are calculated using chain rule

4️⃣ Weight Update

Weights are updated using gradient descent

New weight=Old weight−Learning rate×Gradient

Loss function → measures error

Gradient → direction of error change

Learning rate → how fast weights change

Chain rule → used to compute gradients